CPU Self-Test Controller (STC) Module

This chapter describes the basics and configuration of the CPU self-test controller present in the device.

Topic		Page

8.1 General Description

The CPU self-test controller (STC) is used to test the ARM-CPU core using the Deterministic Logic Built-in Self-Test (LBIST) Controller as the test engine. To achieve better coverage for the self-test of complex cores like Cortex-R4, on-chip logic BIST is the preferred solution.

8.1.1 CPU Self-Test Controller Features

The CPU self-test controller has the following features:

- Capable of running the complete test as well as running a few intervals at a time
  – Ability to continue from the last executed interval (test set) as well as the ability to restart from the beginning (first test set)
  – Total of 24 intervals supported in this device
- Complete isolation of the self-tested CPU core from the rest of the system during the self-test run
  – The self-tested CPU core master bus transaction signals are configured to be in idle mode during the self-test run
  – Any master access to the CPU core under self-test (example: DMA access to CPU TCM) will be held until the completion of the self-test
- Ability to capture the failure interval number
- Timeout counter for the CPU self-test run as a fail-safe feature
- Able to read the MISR data (shifted from LBIST controller) of the last executed interval of the self-test run for debugging purposes
- STCCLK determines the self-test execution speed, STC clock divider (STCCLKDIV) register in the system module is used to divide HCLK (system clock) to generate STCCLK

8.1.2 STC Block Diagram

STC module provides an interface to the LBIST controller implemented on the core.

The CPU STC is composed of following blocks of logic:

- ROM Interface
- FSM and Sequence Control
- Register Block
- Peripheral Bus Interface (VBUSP Interface)
- STC Bypass/ATE Interface

8.1.2.1 ROM Interface

This block handles the ROM address and control signal generation to read the self-test microcode from the ROM. The test microcode and golden signature value for each interval are stored in ROM.

8.1.2.1.1 FSM and Sequence Control

This block generates the signals and data to the LBIST controller based on the seed, test_type and scan chain depth.

8.1.2.1.2 Clock Control

The CLOCK CNTRL sub-block handles the internal clock selection and clock generation for the ROM and LBIST controller.

8.1.2.2 Register Block

This block handles the control of the self-test controller. This block contains various configuration and status registers that provide the result of a self-test run. These registers are memory-mapped and accessible through the Peripheral Bus (VBUSP) Interface. This block controls the reseeding (reloading the existing seed of the PRPG) in the LBIST controller.

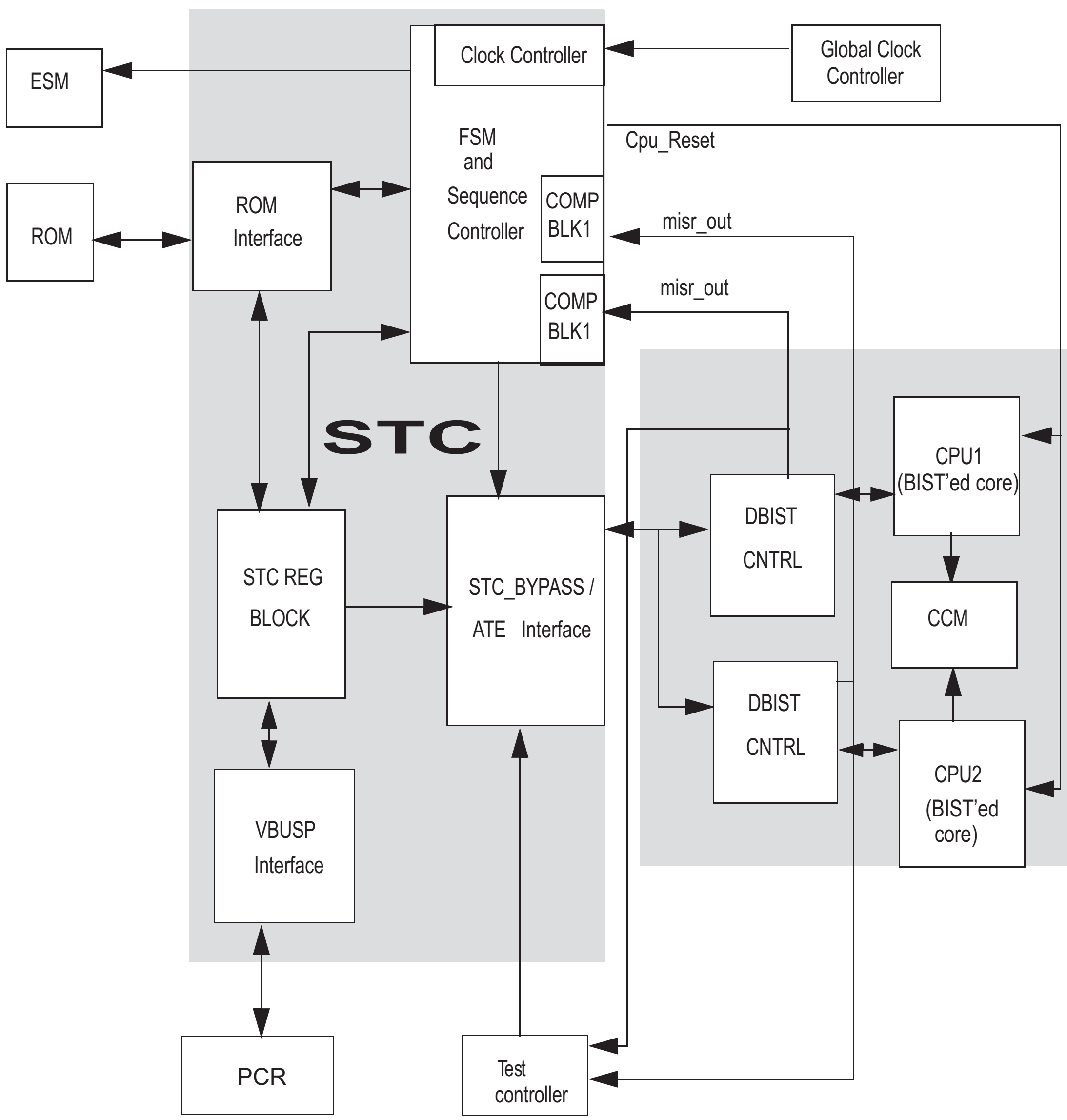
8.1.2.3 STC Bypass / ATE Interface

This is a production test interface. Only for TI internal use.

8.1.2.4 Peripheral Bus (VBUSP) Interface

STC control registers are accessed through Peripheral Bus (VBUSP) Interface. During application programming, configuration registers are programmed through the Peripheral Bus Interface to enable and run the self-test controller.

Figure 8-1. STC Block Diagram



8.2 Application Self-Test Flow

This section describes the STC module configuration and the application self-test flow that the user should follow for successful execution. The following two configurations must be part of the STC initialization code:

- STC clock rate configuration, STC clock divider (STCCLKDIV) register in system module is used to divide HCLK (system clock) to generate STCCLK
- Clear SYSESR register before triggering an STC test

8.2.1 STC Module Configuration

- Configure the test interval count using STCGCR0[31:16] register. A maximum of 24 intervals are supported in the device. You can run 24 intervals together or in slices. If the tests are run in slices, the user software can specify to the self-test controller whether to continue the run from the next interval onwards or to restart from interval 0 using bit STCGCR0[0]. This bit gets reset after the completion of the self-test run.
- Configure self-test run timeout counter preload register STCTPR. This register contains the total number of VBUS clock cycles it will take before a self-test timeout error (TO_ERR) will be triggered after the initiation of the self-test run.
- Enable CPU self-test by writing the enable key to STCGCR1 register.

8.2.2 Context Saving

STC generates a CPU reset after completion of the test regardless of pass or fail. You can run the STC test during startup or can divide STC into 24 or fewer intervals and run them during normal operation.

If STC is ran only on startup, the user software need not save the CPU contents since the reset caused will go through all startup configurations. You should check the STCGSTAT register for the self-test status before going to the application software.

If STC is divided into intervals and ran, user software must save the CPU contents and reload them after the CPU reset caused by the completion of the STC test interval. The check for STC status should bypass STC run if the reset is caused by an STC run to prevent a cyclic reset, that is, if reset is caused by STC the second time through, then it should not be ran again. You should also check the STCGSTAT register for the self-test status before restoring the application software.

Following are some of the registers that are required to be backed up before and restored after self-test:

1. CPU core registers (all modes R0-R15, PC, CPSR)
2. CP15 System Control Coprocessor registers - MPU control and configuration registers, Auxiliary Control Register used to Enable ECC, Fault Status Register etc.
3. CP13 Coprocessor Registers - FPU configuration registers, General Purpose Registers
4. Hardware Break Point and watch point registers like BVR, BSR, WVR, WSR etc.

For more information on the CPU reset, refer to the ARM® Cortex®-R4F Technical Reference Manual.

NOTE: Check all reset source flags in the SYSESR register after a CPU BIST execution. If a flag, in addition to CPU reset, is set, clear the CPU reset flag and service the other reset sources accordingly.

8.2.3 Entering CPU Idle Mode

After enabling the STC test by writing the STC enable key, the test is triggered only after the CPU is taken to idle mode by executing the CPU Idle Instruction **asm(“ WFI”).**

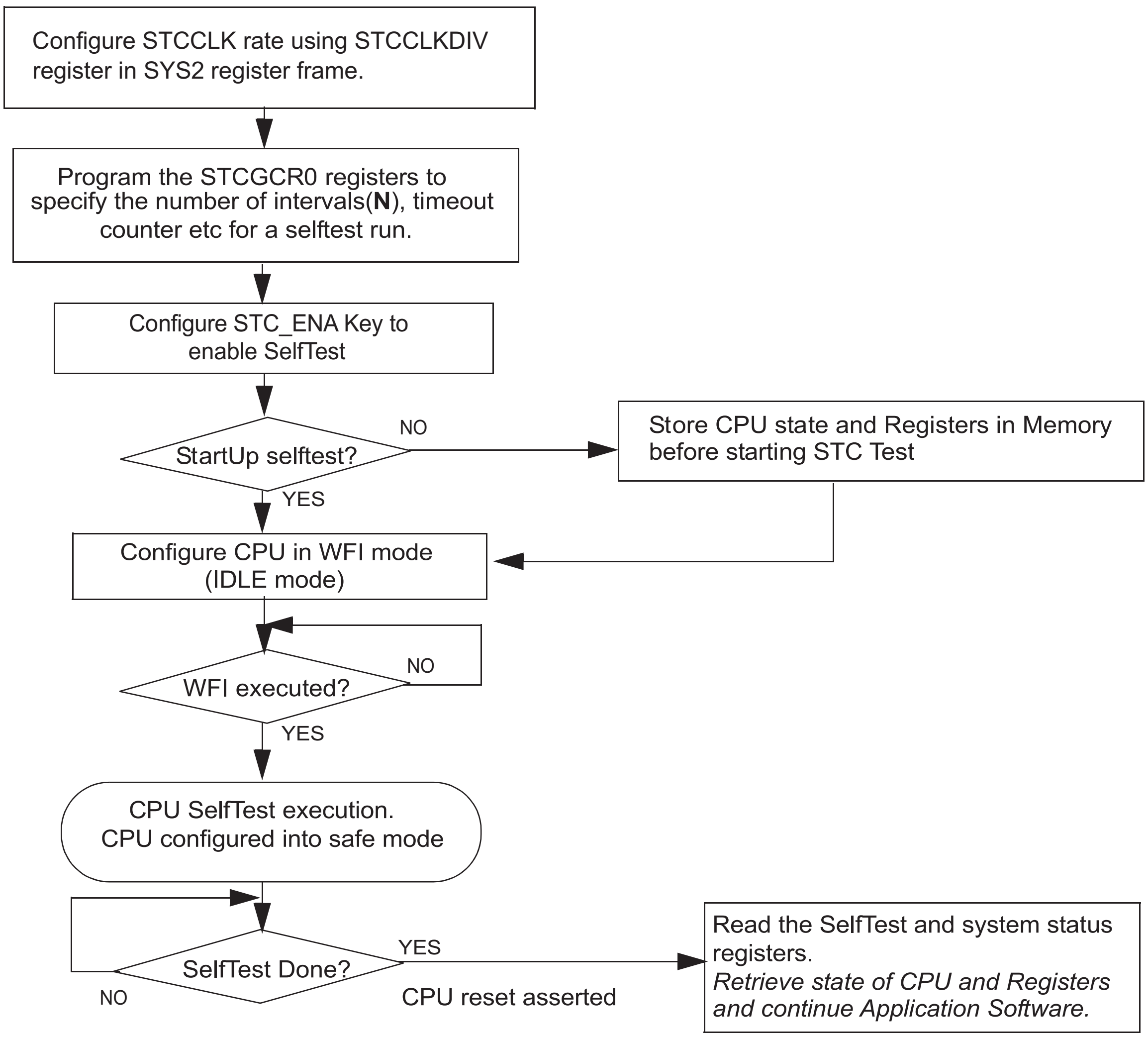
8.2.4 Self-Test Completion and Error Generation

At the end of each interval, the 128 bit MISR value (reflected in registers CPUx_CURMISR[3:0]) from the DBIST controller is shifted into the STC. This is compared with the golden MISR value stored in the ROM.

At the end of a CPU self-test, the STC controller updates the status flags in the Global Status Register (STCGSTAT) and resets the CPU. In case of a MISR mismatch or a test timeout, an error is generated through the ESM module. TEST_ERR signal is asserted when an MISR miss-compare occurs during the self-test. A TO_ERR is asserted when a timeout occurs during the self-test, meaning the test could not complete within the time specified in the timeout counter preload register STCTPR. However, at the device level, these two errors are combined and mapped to a single ESM channel. To identify which error occurred, user software must check the global status register (STCGSTAT) and fail status register STCFSTAT in the ESM interrupt service routine.

Figure 8-2 illustrates the application self-test test flow chart, drawn based on the assumption that the device has gone through startup, necessary clocks initialized and SYSESR register bits cleared.

Figure 8-2. Application Self-Test Flow Chart



8.3 STC Test Coverage and Duration

The test coverage and number of test execution cycles (STCCLK) for each test interval when the device is running at HCLK = 180 MHz, VCLK = 90 MHz, and STCCLK = 90 MHz are shown in Table 8-1.

Table 8-1. STC Test Coverage and Duration

Intervals	Test Coverage (%)	Test Time (Cycles)	Test Time (µs)
0	0	0	0
1	62.13	1365	15.17
2	70.09	2730	30.33
3	74.49	4095	45.50
4	77.28	5460	60.67
5	79.28	6825	75.83
6	80.90	8190	91.00
7	82.02	9555	106.17
8	83.10	10920	121.33
9	84.08	12285	136.50
10	84.87	13650	151.67
11	85.59	15015	166.83
12	86.11	16380	182.00
13	86.67	17745	197.17
14	87.16	19110	212.33
15	87.61	20475	227.50
16	87.98	21840	242.67
17	88.38	23205	257.83
18	88.69	24570	273.00
19	88.98	25935	288.17
20	89.28	27300	303.33
21	89.50	28665	318.50
22	89.76	30030	333.67
23	90.01	31395	348.83
24	90.21	32760	364.00

Table 8-2 gives the typical STC execution times for 24 intervals at different clock rates.

Table 8-2. Typical STC Execution Times

Number of Intervals	@ HCLK = 180 MHz VCLK = 90 MHz STCCLK = 90 MHz	@ HCLK = 100 MHz VCLK = 100 MHz STCCLK = 50MHz	@ HCLK = 160 MHz VCLK = 80 MHz STCCLK = 80 MHz
24	364 µs	655.20 µs	409.50 µs

8.4 STC Control Registers

STC control registers are accessed through Peripheral Bus (VBUSP) interface. Read and write access in 8, 16, and 32 bit are supported. The base address for the control registers is FFFF E600h.

NOTE: In suspend mode, all registers can be written irrespective of user or privilege mode.

Table 8-3. STC Control Registers

Offset	Acronym	Register Description	Section
00h	STCGCR0	STC Global Control Register 0	Section 8.4.1
04h	STCGCR1	STC Global Control Register 1	Section 8.4.2
08h	STCTPR	Self-Test Run Timeout Counter Preload Register	Section 8.4.3
0Ch	STC_CADDR	STC Current ROM Address Register	Section 8.4.4
10h	STCCICR	STC Current Interval Count Register	Section 8.4.5
14h	STCGSTAT	Self-Test Global Status Register	Section 8.4.6
18h	STCFSTAT	Self-Test Fail Status Register	Section 8.4.7
1Ch	CPU1_CURMISR3	CPU1 Current MISR Register	Section 8.4.8
20h	CPU1_CURMISR2	CPU1 Current MISR Register	Section 8.4.8
24h	CPU1_CURMISR1	CPU1 Current MISR Register	Section 8.4.8
28h	CPU1_CURMISR0	CPU1 Current MISR Register	Section 8.4.8
2Ch	CPU2_CURMISR3	CPU2 Current MISR Register	Section 8.4.9
30h	CPU2_CURMISR2	CPU2 Current MISR Register	Section 8.4.9
34h	CPU2_CURMISR1	CPU2 Current MISR Register	Section 8.4.9
38h	CPU2_CURMISR0	CPU2 Current MISR Register	Section 8.4.9
3Ch	STCSCSCR	Signature Compare Self-Check Register	Section 8.4.10

8.4.1 STC Global Control Register 0 (STCGCR0)

This register is described in Figure 8-3 and Table 8-4.

Figure 8-3. STC Global Control Register 0 (STCGCR0) [offset = 00]

31–16
INTCOUNT
R/W-1

15–1	0
Reserved	RS_CNT
R-0	R/WP-0

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -*n* = value after reset

Table 8-4. STC Global Control Register 0 (STCGCR0) Field Descriptions

Bit	Field	Value	Description
31-16	INTCOUNT		Number of intervals of self-test run This register specifies the number of intervals to run for the self-test run. This corresponds to the number of intervals to be ran from the value reflected in the current interval counter.
15-1	Reserved	0	Read returns 0. Writes have no effect.
0	RS_CNT		Restart or Continue This bit specifies whether to continue the run from next interval onwards or to restart from interval 0. This bit gets reset after the completion of a self-test run.
		0	Continue STC run from the previous interval.
		1	Restart STC run from interval 0.

NOTE: On a power-on reset or system reset, this register gets reset to its default values.

8.4.2 STC Global Control Register 1 (STCGCR1)

This register is described in Figure 8-4 and Table 8-5.

Figure 8-4. STC Global Control Register 1 (STCGCR1) [offset = 04h]

31–16
Reserved
R-0

15–4	3–0
Reserved	STC_ENA
R-0	R/WP-5h

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -*n* = value after nPORST (power-on reset) or System reset

Table 8-5. STC Global Control Register 1 (STCGCR1) Field Descriptions

Bit	Field	Value	Description
31-4	Reserved	0	Read returns 0. Writes have no effect.
3-0	STC_ENA		Self-test run enable key
		Ah	Self-test run is enabled.
		All Others	Self-test run is disabled.

NOTE: On a power-on reset or system reset, this register resets to its default values. Also, this register automatically resets to its default values at the completion of a self-test run.

8.4.3 Self-Test Run Timeout Counter Preload Register (STCTPR)

This register is described in Figure 8-5 and Table 8-6.

Figure 8-5. Self-Test Run Timeout Counter Preload Register (STCTPR) [offset = 08h]

31	16
RTOD	
R/WP-FFFFh	

15	0
RTOD	
R/WP-FFFFh	

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -*n* = value after nPORST (power-on reset) or System reset

Table 8-6. Self-Test Run Timeout Counter Preload Register (STCTPR)

Bit	Field	Description
31-0	RTOD	Self-test timeout count preload This register contains the total number of VBUS clock cycles it will take before an self-test timeout error (TO_ERR) will be triggered after the initiation of the self-test run. This is a fail safe feature to prevent the device from hanging up due to a run away test during the self-test. The preload count value gets loaded into the self-test time out down counter whenever a self-test run is initiated (STC_KEY is enabled) and gets disabled on completion of a self-test run.

NOTE: On a power-on reset or system reset, this register gets reset to its default values.

8.4.4 STC Current ROM Address Register (STC_CADDR)

This register is described in Figure 8-6 and Table 8-7.

Figure 8-6. STC Current ROM Address Register (STC_CADDR) [offset = 0Ch]

31	16
ADDR	
R-0	

15	0
ADDR	
R-0	

LEGEND: R/W = Read/Write; R = Read only; -*n* = value after nPORST (power-on reset) or System reset

Table 8-7. STC Current ROM Address Register (STC_CADDR) Field Descriptions

Bit	Field	Description
31-0	ADDR	Current ROM Address This register reflects the current ROM address (for micro code load) which is the current value of the STC program counter.

NOTE: When the RS_CNT bit in STCGCR0 is set to a 1 on the start of a self-test run, or on a power-on reset or system reset, this register resets to all zeroes.

8.4.5 STC Current Interval Count Register (STCCICR)

This register is described in Figure 8-7 and Table 8-8.

Figure 8-7. STC Current Interval Count Register (STCCICR) [offset = 10h]

31	16
Reserved	
R-0	

15	0
N	
R-0	

LEGEND: R/W = Read/Write; R = Read only; -*n* = value after reset

Table 8-8. STC Current Interval Count Register (STCCICR) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Read returns 0. Writes have no effect.
15-0	N		Interval Number This specifies the last executed interval number.

NOTE: When the RS_CNT bit in STCGCR0 is set to a 1 or on a power-on reset, the current interval counter resets to the default value.

8.4.6 Self-Test Global Status Register (STCGSTAT)

This register is described in Figure 8-8 and Table 8-9.

Figure 8-8. Self-Test Global Status Register (STCGSTAT) [offset = 14h]

31–8
Reserved
R-0

31–2	1	0
Reserved	TEST_FAIL	TEST_DONE
R-0	R/WP-0	R/WP-0

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -*n* = value after reset

Table 8-9. Self-Test Global Status Register (STCGSTAT) Field Descriptions

Bit	Field	Value	Description
31-2	Reserved	0	Read returns 0. Writes have no effect.
1	TEST_FAIL		Test Fail
		0	Self-test run has not failed.
		1	Self-test run has failed.
0	TEST_DONE		Test Done
		0	Not completed.
		1	Self-test run completed.
			The test done flag is set to a 1 for any of the following conditions: 1. When the STC run is complete without any failure 2. When a failure occurs on a STC run 3. When a timeout failure occurs
			Reset is generated to the CPU on which the STC run is being performed when TEST_DONE goes high (the test is completed).

NOTE: The two status bits can be cleared to their default values on a write of 1 to the bits. Additionally when the STC_ENA key is written from a disabled state to enabled state, the two status flags get cleared to their default values. This register gets reset to its default value with power-on reset assertion.

8.4.7 Self-Test Fail Status Register (STCFSTAT)

This register is described in Figure 8-9 and Table 8-10.

Figure 8-9. Self-Test Fail Status Register (STCFSTAT) [offset = 18h]

31–8
Reserved
R-0

31–3	2	1	0
Reserved	TO_ERR	CPU2_FAIL	CPU1_FAIL
R-0	R/WP-0	R/WP-0	R/WP-0

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -*n* = value after nPORST (power-on reset) or System reset

Table 8-10. Self-Test Fail Status Register (STCFSTAT) Field Descriptions

Bit	Field	Value	Description
31-3	Reserved	0	Read returns 0. Writes have no effect.
2	TO_ERR		Timeout Error
		0	No time out error occurred.
		1	Self-test run failed due to a timeout error.
1	CPU2_FAIL		CPU2 failure info
		0	No MISR mismatch for CPU2.
		1	Self-test run failed due to MISR mismatch for CPU2.
0	CPU1_FAIL		CPU1 failure info
		0	No MISR mismatch for CPU1.
		1	Self-test run failed due to MISR mismatch for CPU1.

NOTE: The three status bits can be cleared to their default values on a write of 1 to the bits. Additionally when the STC_ENA key in STCGCR1 is written from a disabled state to an enabled state, the three status bits get cleared to their default values. This register gets reset to its default value with power-on reset assertion.

8.4.8 CPU1 Current MISR Register (CPU1_CURMISR[3:0])

This register is described in Figure 8-10 through Figure 8-13 and Table 8-11.

Figure 8-10. CPU1 Current MISR Register (CPU1_CURMISR3) [offset = 1Ch]

31	16
MISR[31:16]	
R-0	

15	0
MISR[15:0]	
R-0	

LEGEND: R/W = Read/Write; R = Read only; -*n* = value after reset

Figure 8-11. CPU1 Current MISR Register (CPU1_CURMISR2) [offset = 20h]

31	16
MISR[63:48]	
R-0	

15	0
MISR[47:32]	
R-0	

LEGEND: R/W = Read/Write; R = Read only; -*n* = value after reset

Figure 8-12. CPU1 Current MISR Register (CPU1_CURMISR1) [offset = 24h]

31	16
MISR[95:80]	
R-0	

15	0
MISR[79:64]	
R-0	

LEGEND: R/W = Read/Write; R = Read only; -*n* = value after reset

Figure 8-13. CPU1 Current MISR Register (CPU1_CURMISR0) [offset = 28h]

31	16
MISR[127:112]	
R-0	

15	0
MISR[111:96]	
R-0	

LEGEND: R/W = Read/Write; R = Read only; -*n* = value after reset

Table 8-11. CPU1 Current MISR Register (CPU1_CURMISR[3:0]) Field Descriptions

Bit	Field	Description
127-0	MISR	MISR data from CPU1 This register contains the MISR data from the CPU1 for the most recent interval. This value is compared with the GOLDEN MISR value copied from ROM.

NOTE: This register gets reset to its default value with power-on or system reset assertion.

8.4.9 CPU2_CURMISR[3:0] (CPU2 Current MISR Register)

This register is described in Figure 8-14 through Figure 8-17 and Table 8-12.

Figure 8-14. CPU2 Current MISR Register (CPU2_CURMISR3) [offset = 2Ch]

31–16
MISR[31:16]
R-0

15–0
MISR[15:0]
R-0

LEGEND: R/W = Read/Write; R = Read only; -*n* = value after reset

Figure 8-15. CPU2 Current MISR Register (CPU2_CURMISR2) [offset = 30h]

31–16
MISR[63:48]
R-0

15–0
MISR[47:32]
R-0

LEGEND: R/W = Read/Write; R = Read only; -*n* = value after reset

Figure 8-16. CPU2 Current MISR Register (CPU2_CURMISR1) [offset = 34h]

31–16
MISR[95:80]
R-0

15–0
MISR[79:64]
R-0

LEGEND: R/W = Read/Write; R = Read only; -*n* = value after reset

Figure 8-17. CPU2 Current MISR Register (CPU2_CURMISR0) [offset = 38h]

31–16
MISR[127:112]
R-0

15–0
MISR[111:96]
R-0

LEGEND: R/W = Read/Write; R = Read only; -*n* = value after reset

Table 8-12. CPU2 Current MISR Register (CPU2_CURMISR[3:0]) Field Descriptions

Bit	Field	Description
127-0	MISR	MISR data from CPU2 This register contains the MISR data from the CPU2 for the most recent interval. This value is compared with the GOLDEN MISR value copied from ROM.

NOTE: This register gets reset to its default value with power-on or system reset assertion.

8.4.10 STCSCSCR (Signature Compare Self-Check Register)

This register is described in Figure 8-18. This register is used to enable the self-check feature of the CPU Self-Test Controller's (STC) signature compare logic. Self-check can only be done for the STC interval 0 by setting the RS_CNT bit in STCGCR0 to 1 to restart the self-test. The STC run will fail for signature miss-compare, provided the signature compare logic is operating correctly. To proceed with regular CPU self-test, STCSCSCR should be programmed to disable the self-check feature and clear the RS_CNT bit in STCGCR0 to 0. This register gets reset to its default value with any system reset assertion.

Figure 8-18. Signature Compare Self-Check Register (STCSCSCR) [offset = 3Ch]

31–16
Reserved
R-0

15–5	4	3–0
Reserved	FAULT_INS	SELF_CHECK_KEY
R-0	R/WP-0	R/WP-5h

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -*n* = value after nPORST (power-on reset) or System reset

Table 8-13. Signature Compare Self-Check Regsiter (STCSCSCR) Field Descriptions

Bit	Field	Value	Description
31-5	Reserved		Reads return zeros, writes have no effect
4	FAULT_INS		Enable / Disable fault insertion.
		0	No fault is inserted.
		1	Insert stuck-at-fault inside CPU so that STC signature compare will fail.
3-0	SELF_CHECK_KEY		Signature compare logic self-check enable key
		Ah	Signature compare logic self-check is enabled. This allows a fault to be inserted using the FAULT_INS field.
		Any other value	Signature compare logic self-check is disabled The FAULT_INS field has no effect in this case.

8.5 STC Configuration Example

The following examples assume that the PLL is locked and selected as the system clock source with HCLK = 180 MHz and VCLK = 90 MHz.

8.5.1 *Example 1: Self-Test Run for 24 Interval*

This example explains the configurations for running STC Test for maximum Test Intervals 24.

1. Maximum STC clock rate support at 180 MHz HCLK is 90 MHz. Divide HCLK by 2 to achieve this clock rate. STCCLKDIV[26:24] register in the secondary system module frame at location 0xFFFF E108 is used.

   STCCLKDIV[26:24] = 1
2. Clear CPU_RST status bit in the System Exception Status Register in the system module.

   SYSESR[5] = 1
3. Configure the test interval count in STC module.

   STCGCR0[31:16] = 24
4. Configure self-test run time out counter preload register.

   STCTPR[31:0] = 0xFFFFFFFF
5. Enable CPU self-test.

   STCGCR1[3:0]= 0xA
6. Perform a context save of CPU state and configuration registers that get reset on CPU reset.
7. Put the CPU in idle mode by executing the CPU idle instruction.
   asm(“ WFI”)
8. Upon CPU reset, verify the CPU_RST status bit in the System Exception Status Register is set. This also verifies that no other resets occurred during the self-test.

   SYSESR[5] == 1
9. Check the STCGSTAT register for the self-test status.

   Check TEST_DONE bit before evaluating TEST_FAIL bit.

   If TEST_DONE = 0 the self-test is not completed. restart the STC test by going to Step 5.

   If (TEST_DONE = 1 and TEST_FAIL = 1) the self-test is completed and Failed.

   - Read STC Fail Status Register STCFSTAT[2:0] to identify the type of Failure (Timeout, CPU1 fail, CPU2 fail).

   In case there is no failure (TEST_DONE = 1 and TEST_FAIL = 0), the CPU self-test is completed successfully.

   - Recover the CPU status, configuration registers and continue the application software.